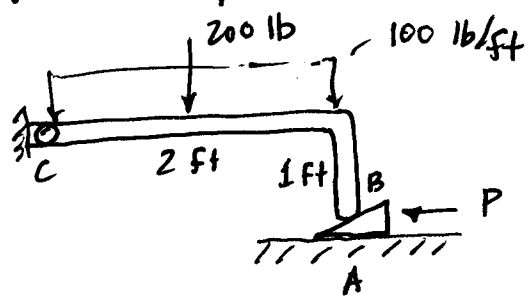


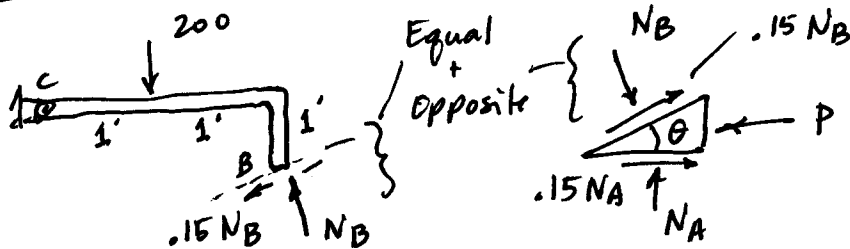
Wedge Example Problem : Raising Case



Wedge: $\theta = 10^\circ$

$$\mu_A = \mu_B = .15$$

Find P to raise the frame member.



} Draw μN terms to oppose motion of wedge

Member CB: $+\circlearrowleft \sum M_C = 0$; $200(1) - N_B \cos 10 (2)$

$$+ N_B \sin 10 (1) + .15 N_B \cos 10 (1) + .15 N_B \sin 10 (2) = 0$$

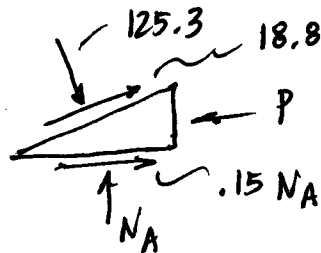
$$200 = N_B (2 \cos 10 - \sin 10 - .15 \cos 10 - .3 \sin 10)$$

$$200 = N_B (1.596)$$

$$N_B = 125.3 \text{ lb}$$

$$.15 N_B = 18.80 \text{ lb}$$

Wedge



$$+\uparrow N_A - 125.3 \cos 10 + 18.8 \sin 10 = 0$$

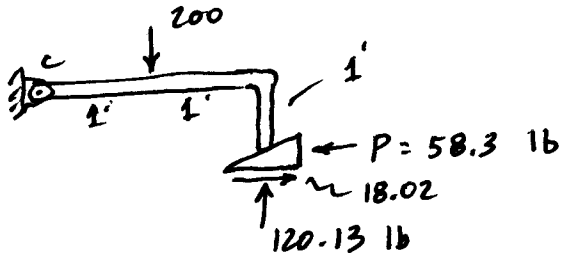
$$N_A = 120.13 \text{ lb}$$

$$.15 N_A = 18.02 \text{ lb}$$

$$+\rightarrow P - 18.02 - 18.8 \cos 10 - 125.3 \sin 10$$

$$P = 58.3 \text{ lb}$$

Check Answers



$$+ (\sum M_c = 0 ; \quad 200(1) + 58.3(1) - 18.02(1) - 120.13(2) = 0$$

yes!
Checks!